

Problem

The primary current to an ideal transformer rated at 2200/110 V is 25 A. Calculate: (a) the turns ratio, (b) the kVA rating, (c) the secondary current.

Step-by-step solution

Step 1 of 3

(a)

Calculate the value of Turns ratio n .

Write the expression for Turns ratio n .

$$\begin{aligned} n &= \frac{V_2}{V_1} \\ &= \frac{110}{2200} \left[\text{Since, } \frac{V_1}{V_2} = \frac{2200 \text{ V}}{110 \text{ V}} \right] \\ &= \frac{1}{20} \end{aligned}$$

Turns ration of the transformer is $\boxed{\frac{1}{20}}$.

Step 2 of 3

(b)

Calculate the value of the KVA rating S of the transformer.

Write the expression for KVA rating S of the transformer.

$$\begin{aligned} S &= V_1 I_1 \\ &= 2200(5) \\ &= 11000 \text{ VA} \end{aligned}$$

Therefore, the value of the KVA rating of the transformer is $\boxed{11 \text{ kVA}}$.

Step 3 of 3

(c)

Calculate the value of the secondary current I_2 .

Write the expression for KVA rating S of the transformer.

$$\begin{aligned} S &= V_2 I_2 \\ I_2 &= \frac{S}{V_2} \\ &= \frac{11000}{110} \\ &= 100 \text{ A} \end{aligned}$$

Therefore, the value of the Secondary current I_2 is $\boxed{100 \text{ A}}$.